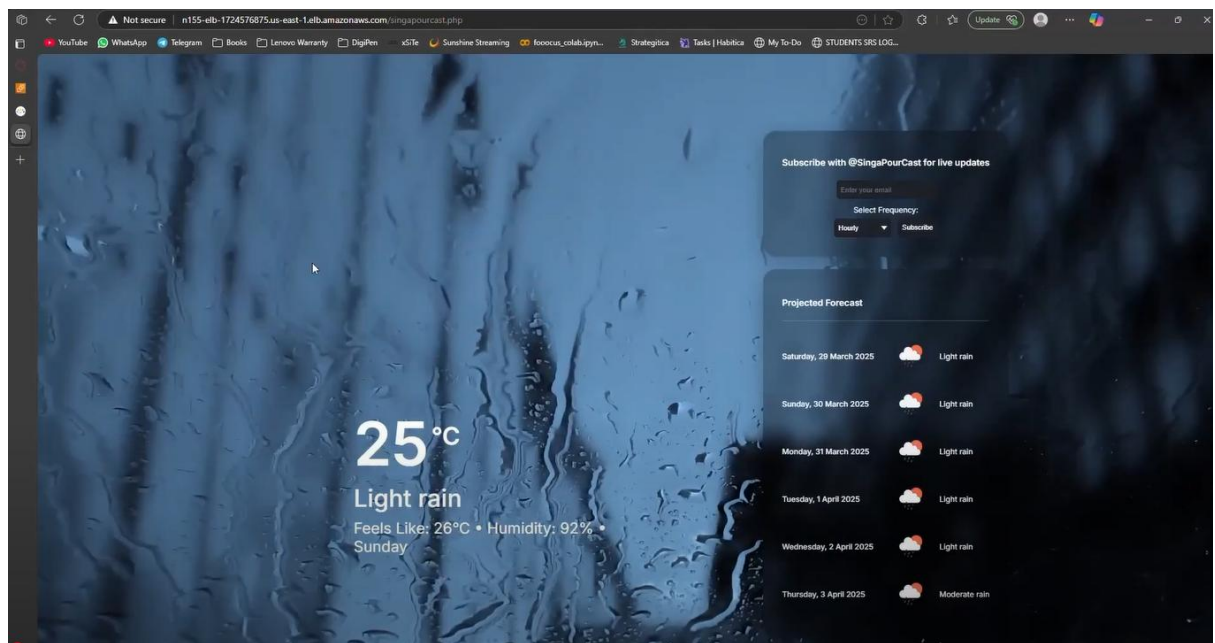


Project Report – Cloud Computing

SingaPourCast

1. Introduction

SingaPourCast is a live Singapore weather tracker web application hosted over AWS. It is designed to provide weather forecasting services by integrating with the OpenWeather API.



The goal of this project is to build a scalable, reliable, and secure cloud-based system that efficiently handles incoming weather requests and delivers forecasts to users while ensuring high availability and performance.

The scope of this project includes designing and deploying a cloud infrastructure using AWS services such as the Elastic Load Balancer (ELB), Auto Scaling Group (ASG), Virtual Private Cloud (VPC), Relational Database Service (RDS) and CloudWatch. This project also involves integrating services outside of AWS such as Gmail SMTP for email notifications and OpenWeather API for the weather forecast.

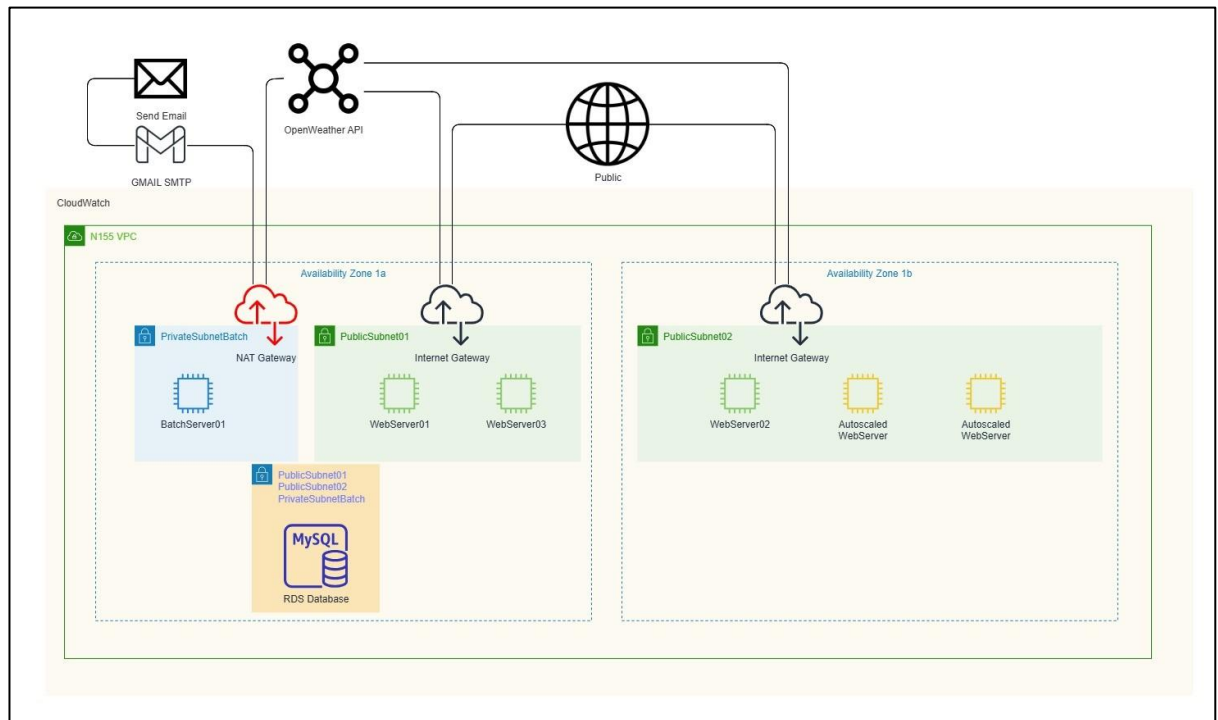
The project was a collaborative effort involving multiple team members, each responsible for different aspects of the project including but not limited to:

- Architecture Design
- Security Enforcement
- Database Management
- Load Balancing
- Frontend Development
- Backend Development
- Third-Party Integration (API & SMTP)
- Stress Testing

2. Architecture Design and Solutions

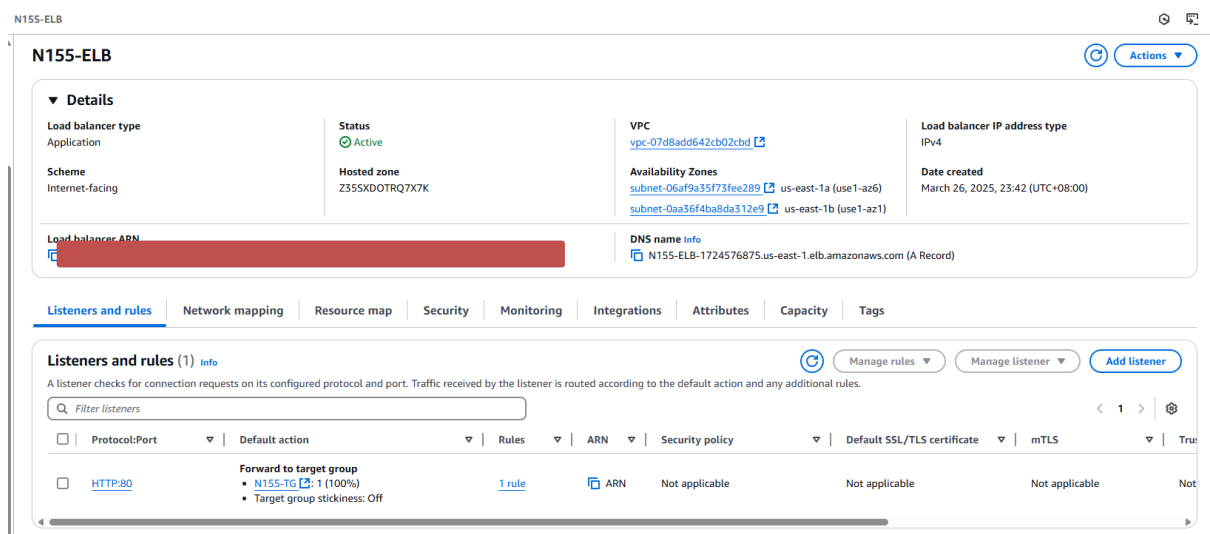
System Architecture

The architecture consists of multiple components working together to ensure smooth operation and deployment. The image below clearly shows the architecture design in a broad sense:



The key components include:

- Elastic Load Balancer (ELB): Distributes traffic across the five web servers to enhance availability and reliability. This ensures at least an instance of the web server is up in the event that one or some shuts down.



N155-TG

N155-TG

Details

Target type Instance	Protocol - Port HTTP: 80	Protocol version HTTP1	VPC vpc-07d8add642cb02cbd
IP address type IPv4	Load balancer N155-ELB		

5 Total targets

5 Healthy
0 Anomalous

0 Unhealthy

0 Unused

0 Initial

0 Draining

Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.

Targets | Monitoring | Health checks | Attributes | Tags

Registered targets (5) [info](#) [Anomaly mitigation: Not applicable](#) [Deregister](#) [Register targets](#)

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

Filter targets

☐	Instance ID	Name	Port	Zone	Health status	Health status details	Administrative ...	Override details	Launch...	Anomaly detection
☐	i-07f0223cc05f1f3d2a		80	us-east-1a (us-...)	Healthy	-	No override	No override is curr...	March 28,...	Normal
☐	i-09982391c522f3eaa		80	us-east-1b (us-...)	Healthy	-	No override	No override is curr...	March 28,...	Normal
☐	i-02aa8b8de54a93683	N155-WebSer...	80	us-east-1a (us-...)	Healthy	-	No override	No override is curr...	March 28,...	Normal
☐	i-03ff6c8d53c3b678	N155-WebSer...	80	us-east-1a (us-...)	Healthy	-	No override	No override is curr...	March 28,...	Normal
☐	i-0cd1c24e7e187526	N155-WebSer...	80	us-east-1b (us-...)	Healthy	-	No override	No override is curr...	March 28,...	Normal

- Auto Scaling Group (ASG): Ensures Scalability by dynamically downsizing or upsizing the number of web servers based on traffic demand.

N155-ASG

N155-ASG Capacity overview

Desired capacity
2

Scaling limits (Min - Max)
2 - 6

Desired capacity type
Units (number of instances)

Status
-

Date created
Thu Mar 27 2025 05:20:00 GMT+0800 (Singapore Standard Time)

Details | Integrations - new | Automatic scaling | Instance management | Instance refresh | Activity | Monitoring

Launch template

Launch template
[lt-0e4654e5ab7977ae6](#)
N155-WebServer02-ASG-Template

Version
Default

Description
-

[View details in the launch template console](#)

AMI ID
[ami-06d2e5b1cc0f97b36](#)

Security groups
-

Storage (volumes)
-

Instance type
t2.micro

Security group IDs
[sg-0feb5e50f536109c9](#)

Key pair name
vockey

Owner
-

Create time
Fri Mar 28 2025 01:54:18 GMT+0800 (Singapore Standard Time)

Request Spot Instances
No

Network

Availability Zones
us-east-1a, us-east-1b

Subnet ID
subnet-06af9a35f73fee289, subnet-0aa36f4ba8da312e9

Availability Zone distribution
Balanced best effort

- Subnets: Web Servers reside in the public subnet, while backend servers like the Batch Server and RDS database are placed in private subnets for security.

Subnets (4) [info](#)

Find resources by attribute or tag

[n155](#) [Clear filters](#)

Last updated less than a minute ago [Actions](#) [Create subnet](#)

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR
☐	N155-PublicSubnet-01	subnet-06af9a35f73fee289	Available	vpc-07d8add642cb02cbd N155	☐ Off	10.0.1.0/24	-
☐	N155-PrivateSubnet-RDS	subnet-0d65f0e0e7f8bea36	Available	vpc-07d8add642cb02cbd N155	☐ Off	10.0.4.0/24	-
☐	N155-PrivateSubnet-Batch	subnet-0579bffd5e01abd4	Available	vpc-07d8add642cb02cbd N155	☐ Off	10.0.3.0/24	-
☐	N155-PublicSubnet-02	subnet-0aa36f4ba8da312e9	Available	vpc-07d8add642cb02cbd N155	☐ Off	10.0.2.0/24	-

subnet-06af9a35f73fee289

subnet-06af9a35f73fee289 / N155-PublicSubnet-01

Actions

Details

Subnet ID
subnet-06af9a35f73fee289

IPv4 CIDR
10.0.1.0/24

Availability Zone
us-east-1a

Route table
rtb-0e5420d3529b88974 | N155-PublicRouteTable

Auto-assign IPv6 address
No

IPv4 CIDR reservations
-

Resource name DNS A record
Disabled

Subnet ARN

Available IPv4 addresses
245

Availability Zone ID
use1-az6

Network ACL
acl-0143dad30dbb6165c

Auto-assign customer-owned IPv4 address
No

IPv6 CIDR reservations
-

Resource name DNS AAAA record
Disabled

State
Available

IPv6 CIDR
-

Network border group
us-east-1

Default subnet
No

Customer-owned IPv4 pool
-

IPv6-only
No

DNS64
Disabled

Block Public Access
Off

IPv6 CIDR association ID
-

VPC
vpc-07d8add642cb02cbd | N155

Auto-assign public IPv4 address
Yes

Outpost ID
-

Hostname type
IP name

Owner

Flow logs Route table Network ACL CIDR reservations Sharing Tags

Route table: rtb-0e5420d3529b88974 / N155-PublicRouteTable

Edit route table association

Routes (2)

Filter routes

Destination

10.0.0.0/16

0.0.0.0/0

Target

local

igw-0616919da18472470

subnet-0aa36f4ba8da312e9

subnet-0aa36f4ba8da312e9 / N155-PublicSubnet-02

Actions

Details

Subnet ID
subnet-0aa36f4ba8da312e9

IPv4 CIDR
10.0.2.0/24

Availability Zone
us-east-1b

Route table
rtb-0e5420d3529b88974 | N155-PublicRouteTable

Auto-assign IPv6 address
No

IPv4 CIDR reservations
-

Resource name DNS A record
Disabled

Subnet ARN

Available IPv4 addresses
248

Availability Zone ID
use1-az1

Network ACL
acl-0143dad30dbb6165c

Auto-assign customer-owned IPv4 address
No

IPv6 CIDR reservations
-

Resource name DNS AAAA record
Disabled

State
Available

IPv6 CIDR
-

Network border group
us-east-1

Default subnet
No

Customer-owned IPv4 pool
-

IPv6-only
No

DNS64
Disabled

Block Public Access
Off

IPv6 CIDR association ID
-

VPC
vpc-07d8add642cb02cbd | N155

Auto-assign public IPv4 address
Yes

Outpost ID
-

Hostname type
IP name

Owner

Flow logs Route table Network ACL CIDR reservations Sharing Tags

Route table: rtb-0e5420d3529b88974 / N155-PublicRouteTable

Edit route table association

Routes (2)

Filter routes

Destination

10.0.0.0/16

0.0.0.0/0

Target

local

igw-0616919da18472470

subnet-0579bffd5e01abd4

subnet-0579bffd5e01abd4 / N155-PrivateSubnet-Batch

Details

Subnet ID
subnet-0579bffd5e01abd4

IPv4 CIDR
10.0.3.0/24

Availability Zone
us-east-1a

Route table
rtb-0426a73fe50d16528 | N155-PrivateRouteTable

Auto-assign IPv6 address
No

IPv4 CIDR reservations
-

Resource name DNS A record
Disabled

Subnet ARN
[REDACTED]

Available IPv4 addresses
250

Availability Zone ID
use1-az6

Network ACL
acl-0143dad30dbb6165c

Auto-assign customer-owned IPv4 address
No

IPv6 CIDR reservations
-

Resource name DNS AAAA record
Disabled

State
Available

IPv6 CIDR
-

Network border group
us-east-1

Default subnet
No

Customer-owned IPv4 pool
-

IPv6-only
No

DNS64
Disabled

Block Public Access
Off

IPv6 CIDR association ID
-

VPC
vpc-07d8add642cb02cbd | N155

Auto-assign public IPv4 address
Yes

Outpost ID
-

Hostname type
IP name

Flow logs | **Route table** | Network ACL | CIDR reservations | Sharing | Tags

Route table: rtb-0426a73fe50d16528 / N155-PrivateRouteTable

Routes (2)

Filter routes

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	nat-0721d2d6490184d86

ps > n155-rds-subnet

n155-rds-subnet

Subnet group details

VPC ID

vpc-07d8add642cb02cbd

ARN

Supported network types

IPv4

Description

N155 Database Subnet

Subnets (3)

Availability zone	Subnet name	Subnet ID	CIDR block
us-east-1a	N155-PublicSubnet-01	subnet-06af9a35f73fee289	10.0.1.0/24
us-east-1a	N155-PrivateSubnet-Batch	subnet-0579bffd5e01abd4	10.0.3.0/24
us-east-1b	N155-PublicSubnet-02	subnet-0aa36f4ba8da312e9	10.0.2.0/24

- Security Groups: Controls the traffic that is allowed to reach and leave the instances.

Security Groups (4) info

Find resources by attribute or tag

VPC ID = vpc-07d8add642cb02cbd

Clear filters

Export security groups to CSV

Create security group

Name	Security group ID	Security group name	VPC ID	Description	Owner
-	sg-095586650cc004879	N155-SG-BatchServer	vpc-07d8add642cb02cbd	Allows access to RDS Database & API Calls	[REDACTED]
-	sg-0f06d5b90749e19ed	default	vpc-07d8add642cb02cbd	default VPC security group	[REDACTED]
-	sg-0feb5e50f536109c9	N155-SG-WebServer	vpc-07d8add642cb02cbd	Allows Public access for the Web Server	[REDACTED]
-	sg-0a2ac5c51b5e7a239	N155-SG-RDS	vpc-07d8add642cb02cbd	Allow access from Web and Batch Servers	[REDACTED]

sg-0feb5e50f536109c9 - N155-SG-WebServer

sg-0feb5e50f536109c9 - N155-SG-WebServer Actions

Details

Security group name N155-SG-WebServer	Security group ID sg-0feb5e50f536109c9	Description Allows Public access for the Web Server	VPC ID vpc-07d8add642cb02cbbd
Owner [Redacted]	Inbound rules count 3 Permission entries	Outbound rules count 1 Permission entry	

[Inbound rules](#) [Outbound rules](#) [Sharing - new](#) [VPC associations - new](#) [Tags](#)

Inbound rules (3) Manage tags Edit inbound rules

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-0710868d58c35667e	IPv4	HTTP	TCP	80	0.0.0.0/0	-
☐	-	sgr-0cb3060017251f046	IPv4	SSH	TCP	22	0.0.0.0/0	-
☐	-	sgr-0adc0bfb8b10f0574	IPv4	HTTPS	TCP	443	0.0.0.0/0	-

sg-095586650cc004879 - N155-SG-BatchServer

sg-095586650cc004879 - N155-SG-BatchServer Actions

Details

Security group name N155-SG-BatchServer	Security group ID sg-095586650cc004879	Description Allows access to RDS Database & API Calls	VPC ID vpc-07d8add642cb02cbbd
Owner [Redacted]	Inbound rules count 4 Permission entries	Outbound rules count 1 Permission entry	

[Inbound rules](#) [Outbound rules](#) [Sharing - new](#) [VPC associations - new](#) [Tags](#)

Inbound rules (4) Manage tags Edit inbound rules

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-02b987acab20e2b43	-	MySQL/Aurora	TCP	3306	sg-0a2ac5c51b5e7a239...	-
☐	-	sgr-00c1bb72d8852e251	IPv4	SSH	TCP	22	103.6.151.70/32	-
☐	-	sgr-085a9756bebd5125a	-	HTTP	TCP	80	sg-0feb5e50f536109c9...	-
☐	-	sgr-09cf72f91a69b316d	-	HTTPS	TCP	443	sg-0feb5e50f536109c9...	-

sg-0a2ac5c51b5e7a239 - N155-SG-RDS

sg-0a2ac5c51b5e7a239 - N155-SG-RDS Actions

Details

Security group name N155-SG-RDS	Security group ID sg-0a2ac5c51b5e7a239	Description Allow access from Web and Batch Servers	VPC ID vpc-07d8add642cb02cbbd
Owner [Redacted]	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

[Inbound rules](#) [Outbound rules](#) [Sharing - new](#) [VPC associations - new](#) [Tags](#)

Inbound rules (2) Manage tags Edit inbound rules

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-041e9bd8241dfb9cf	-	MySQL/Aurora	TCP	3306	sg-0feb5e50f536109c9...	-
☐	-	sgr-05498ad9f07c95f3c	-	MySQL/Aurora	TCP	3306	sg-095586650cc00487...	-

- Internet (IGW) & NAT Gateway: IGW allows resources in Public Subnets to access internet while receiving traffic while NAT allows resources in Private Subnets to access the internet securely.

igw-0616919da18472470

igw-0616919da18472470 / N155-IGW

Details [info](#)

Internet gateway ID igw-0616919da18472470	State Attached	VPC ID vpc-07d8add642cb02cbd N155	Owner [Redacted]
---	--------------------------	---	----------------------------

Tags [Manage tags](#)

Search tags

Key	Value
Name	N155-IGW

nat-0721d2d6490184d86

nat-0721d2d6490184d86 / N155-PublicNAT

Details

NAT gateway ID nat-0721d2d6490184d86	Connectivity type Public	State Available	State message info -
NAT gateway ARN [Redacted]	Public IPv4 address [Redacted]	Primary private IPv4 address 10.0.1.234	Primary network interface ID eni-0cbb0c7519f9166a info
VPC vpc-07d8add642cb02cbd N155	Subnet subnet-06af9a35f73fee289 N155-PublicSubnet-01	Created Wednesday, March 26, 2025 at 22:42:46 GMT+8	Deleted -

Secondary IPv4 addresses | [Monitoring](#) | [Tags](#)

Search

Private IPv4 address	Allocation ID	Association ID	Public IPv4 address	Network interface ID	Status
Secondary IPv4 addresses are not available for this nat gateway.					

- RDS MySQL: Manages the subscribed email, frequency and next send time.

n155-database

n155-database

[Modify](#) [Actions](#)

Summary

DB identifier n155-database	Status Available	Role Instance	Engine MySQL Community	Recommendations 4 Informational
CPU 3.45%	Class db.t3.micro	Current activity 0 Connections	Region & AZ us-east-1a	

[Connectivity & security](#) | [Monitoring](#) | [Logs & events](#) | [Configuration](#) | [Zero-ETL integrations](#) | [Maintenance & backups](#) | [Data migrations - new](#) | [Tags](#) | [Recommendations](#)

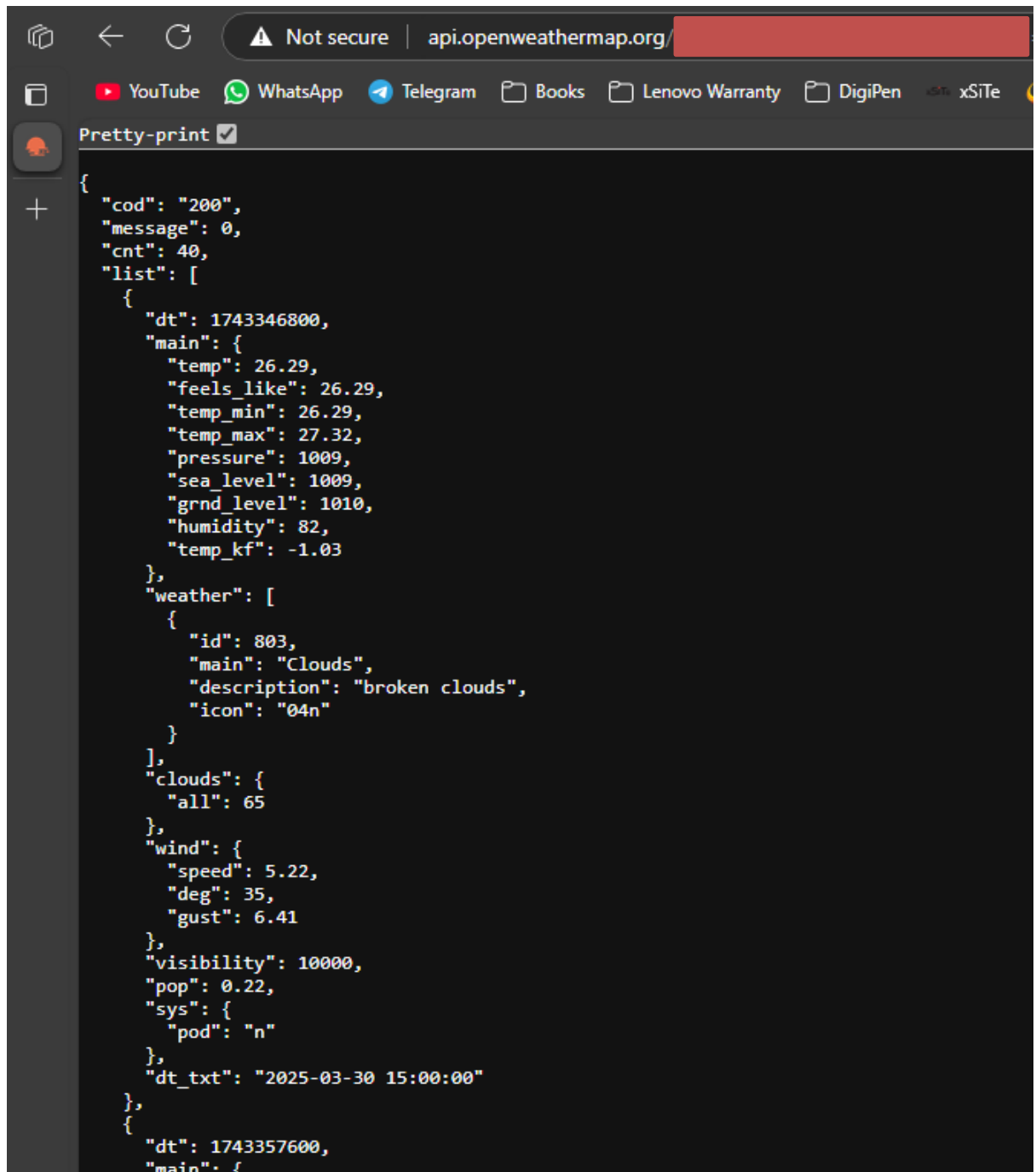
Connectivity & security

Endpoint & port	Networking	Security
Endpoint n155-database.cb4antwaqkle.us-east-1.rds.amazonaws.com	Availability Zone us-east-1a	VPC security groups N155-SG-RDS (sg-0a2ac5c51b5e7a239) Active
Port 3306	VPC N155 (vpc-07d8add642cb02cbd)	Publicly accessible No
	Subnet group n155-rds-subnet	Certificate authority info rds-ca-rsa2048-g1
	Subnets subnet-06af9a35f73fee289 subnet-0aa36f4ba8da312e9 subnet-0579bfddc5e01abd4	Certificate authority date May 26, 2061, 07:34 (UTC+08:00)
	Network type IPv4	DB instance certificate expiration date March 27, 2026, 01:11 (UTC+08:00)

- CloudWatch: Monitors and logs webapp performance and auditing.

Functional Components

- Weather Data Processing: Fetches real-time weather data from OpenWeather API.



The screenshot shows a web browser window with the address bar displaying 'api.openweathermap.org/'. The browser's address bar also shows a 'Not secure' warning. The browser's tab bar includes links to YouTube, WhatsApp, Telegram, Books, Lenovo Warranty, DigiPen, and xSITE. The main content area displays a JSON response from the OpenWeatherMap API, formatted in a 'Pretty-print' view. The JSON data includes a 'cod' of '200', a 'message' of '0', a 'cnt' of '40', and a 'list' of weather objects. The first object in the list shows a timestamp of '1743346800', a temperature of '26.29', and a weather condition of 'broken clouds' with an icon of '04n'. The JSON also includes details about wind, visibility, and humidity.

```
{
  "cod": "200",
  "message": 0,
  "cnt": 40,
  "list": [
    {
      "dt": 1743346800,
      "main": {
        "temp": 26.29,
        "feels_like": 26.29,
        "temp_min": 26.29,
        "temp_max": 27.32,
        "pressure": 1009,
        "sea_level": 1009,
        "grnd_level": 1010,
        "humidity": 82,
        "temp_kf": -1.03
      },
      "weather": [
        {
          "id": 803,
          "main": "Clouds",
          "description": "broken clouds",
          "icon": "04n"
        }
      ],
      "clouds": {
        "all": 65
      },
      "wind": {
        "speed": 5.22,
        "deg": 35,
        "gust": 6.41
      },
      "visibility": 10000,
      "pop": 0.22,
      "sys": {
        "pod": "n"
      },
      "dt_txt": "2025-03-30 15:00:00"
    },
    {
      "dt": 1743357600,
      "main": {
```

- User Notifications: Subscribed Users receive weather alerts via email using Gmail SMTP.



SingaPourCast - Weather Updates <singapourcast@gmail.com>

to me ▾

Weather Forecast:

Date & Time (SGT): 2025-03-30 04:09:02

Temperature: 25.83 °C

Feels Like: 26.87 °C

Condition: light rain

Humidity: 92%

- Web Application Hosting: Users interact with the system through a web interface hosted on the cloud.

Workflow Scenarios

- A user accesses the web application, triggering a request to fetch weather data.
- The request is processed by one of the web servers assigned by the ELB.
- If the user subscribes, the web server stores the 'email', 'frequency' and 'next send time' to the RDS database.
- The batch server then goes through the database and process the job to send email to the subscriber through Composer PHPMailer & Gmail SMTP and updates the 'next send time' of the subscriber to the database.
- CloudWatch monitors performance metrics and logs any unusual activities.

Deployment

- Instances were manually launched on AWS, and all necessary security protocols were configured.
- The team connected to the instances to set up both front-end and back0end scripts.
- PHPMailer was installed using Composer for email notifications in the Batch Server.
- Cron Jobs were configured on the Batch Server to automate scheduled tasks.

3. Discussions

Scalability

- Auto Scaling Groups dynamically add or remove web server instances based on traffic.
- ELB distributes incoming requests across multiple Web Servers.

Reliability

- Multi-AZ deployment and Load Balancing of Web Servers in case of failovers.
- CloudWatch monitoring for real-time performance insights.

Security

- Placement of the RDS Database in a private subnet to restrict direct external access.
- Placement of the Batch Server in a private subnet to restrict direct external access.
- Use of NAT Gateway to allow outbound traffic while keeping Batch Servers protected.
- Web servers are configured with inbound rules allowing only HTTP, HTTPS, and SSH traffic to ensure restricted access.

Limitations

- Since our accounts are restricted and are unable to create IAM roles. This affects using AWS' automation and notification services, leading us to use Composer PHPMailer, Gmail SMTP and Cron Jobs for email notifications and job scheduling respectively.'
- Using Cron Jobs affected multiple Batch Servers job processing, leading to multiple of the same email sent to each subscribers for each notification, thus leading us to using only one Batch Server, increasing risk of functional issues in the event this singular Batch Server goes down.

Improvements

- Integrate third party Job Schedulers like SOSBerlin so that we can have more than one Batch

Server running on a separate ELB to for a more efficient Job Processing, scalability and reliability of the Batch Servers in the events of failovers.

- Set up meaningful alarms in CloudWatch like CPU, Memory and Storage of each instance going over certain thresholds that warrants attention or action.
- Add on more functionalities of the Web Application such as automatic Disaster Alerts, Thunderstorm Alerts, etc. Also ability to integrate with other applications like a calendar app to help users plan their day.

4. Reflection and takeaways

Through the SingaPourCast project, the team gained valuable experience in designing and deploying cloud-based applications. Key learnings include:

- Understanding the importance of well-architected AWS infrastructure for scalability and reliability.
- Implementing security best practices to protect sensitive data.

Future improvements can focus on further optimizing costs, enhancing security, and introducing new features like machine learning-based weather predictions. The project demonstrated the power of cloud computing in delivering robust and efficient applications.

Reference

The references are a list of any sources you have used in your report. Your report should use the standard referencing style following an established referencing format (e.g., IEEE, Harvard, APA, etc.).

- [1] G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," Phil. Trans. Roy. Soc. London, vol. A247, pp. 529–551, April 1955. (references)

END OF REPORT